

Sorting Algorithm Benchmark Report

Insertion Sort vs Selection Sort vs Merge Sort

This report compares measured running times for three sorting algorithms across multiple input patterns and input sizes. The graphs show how runtime changes as n grows, and the tables estimate where merge sort becomes consistently faster.

Benchmark rows

10,521

Input patterns

7

Algorithms

3

Input range

0-500

Report outputs

Individual PNG graphs saved in results/plots

PDF report saved as results/sorting_graph_report.pdf

Crossover summary saved as results/crossover_summary.csv

Executive summary

Stable crossover points by input type

Input type	Insertion vs Merge	Selection vs Merge
Random	n = 106	n = 79
Sorted	No stable crossover	n = 78
Reverse	n = 44	n = 80
Duplicates	n = 114	n = 88
Nearly Sorted	No stable crossover	n = 93
All Equal	No stable crossover	n = 84
Few Unique	n = 125	n = 92

Stable crossover means merge sort was faster for five consecutive input sizes starting from that value of n.

How to read the graphs

Interpretation guide for the benchmark results

X-axis

Input size n . Larger n means the algorithm sorted a longer list.

Y-axis

Average running time in milliseconds. Lower values mean faster execution.

Lines

Each line represents one algorithm. Selection sort appears as its own separate line.

Crossover markers

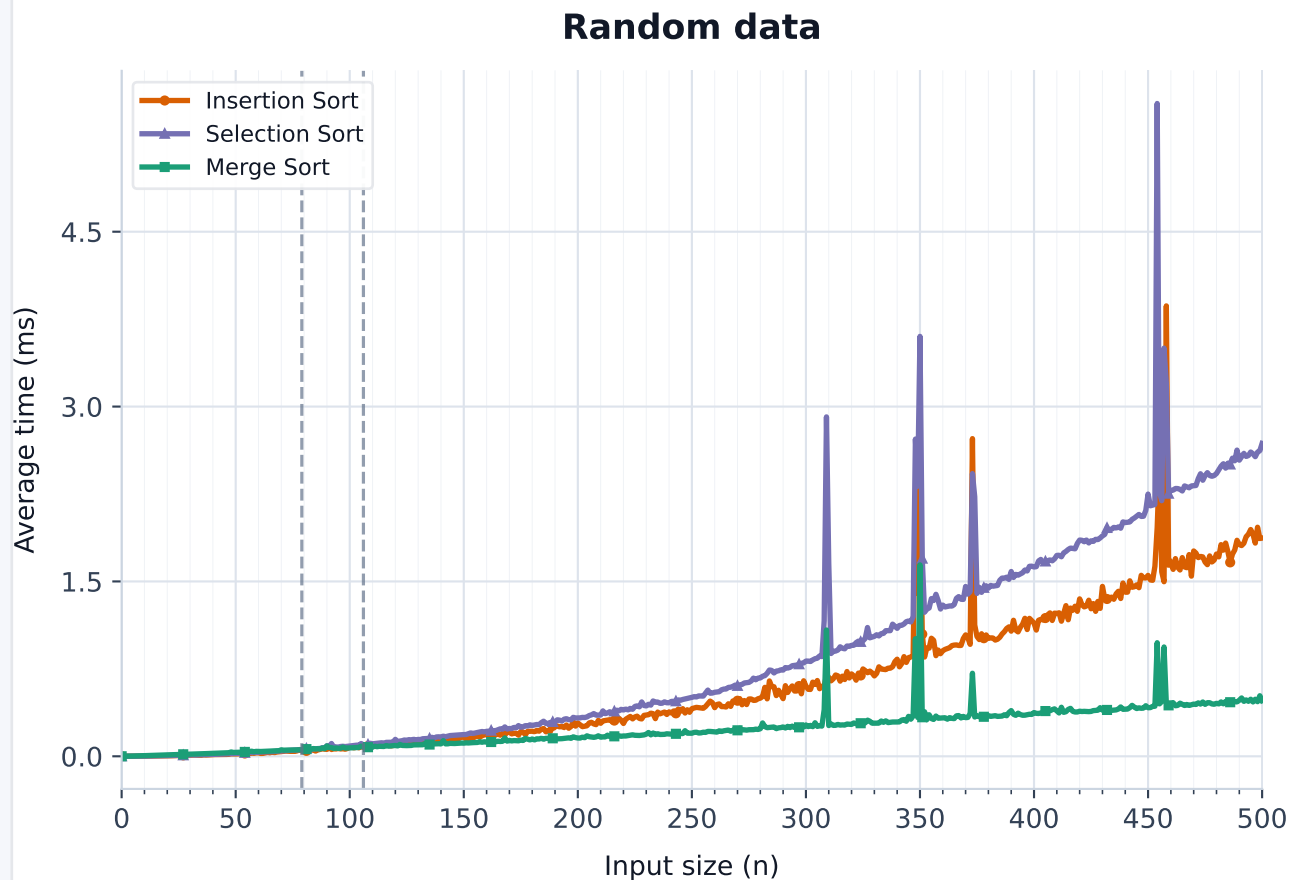
Vertical dashed lines mark where merge sort becomes consistently faster than an $O(n^2)$ algorithm.

Limitation

Small benchmark timings are noisy, so the report uses repeated runs and a consecutive-size rule instead of relying on one isolated point.

Random input

Average running time by input size



Key results

Fastest at largest n

Merge Sort
when $n = 500$

Stable crossovers

Insertion vs Merge
 $n = 106$

Selection vs Merge
 $n = 79$

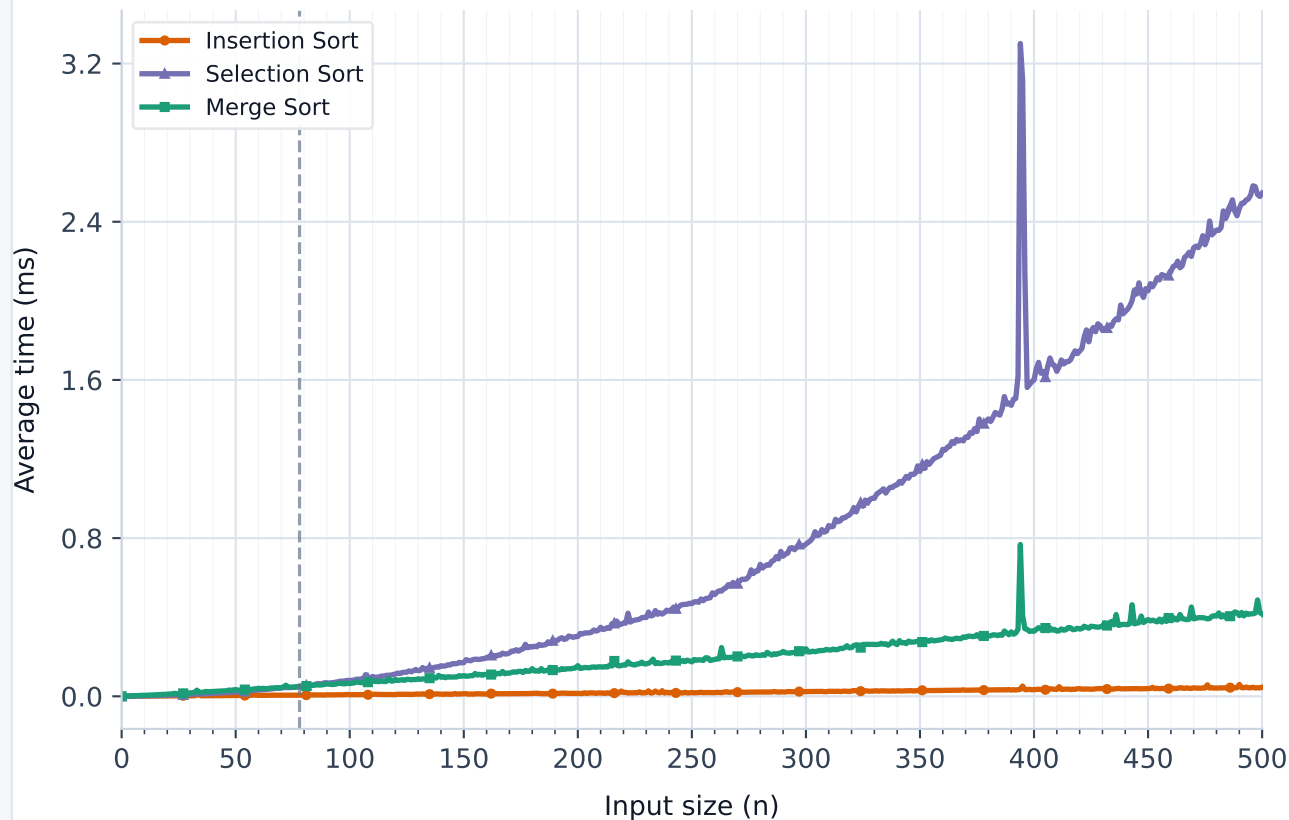
Reading note

Lower line means faster.
Dashed line means stable crossover.

Sorted input

Average running time by input size

Sorted data



Key results

Fastest at largest n

Insertion Sort
when $n = 500$

Stable crossovers

Insertion vs Merge
No stable crossover

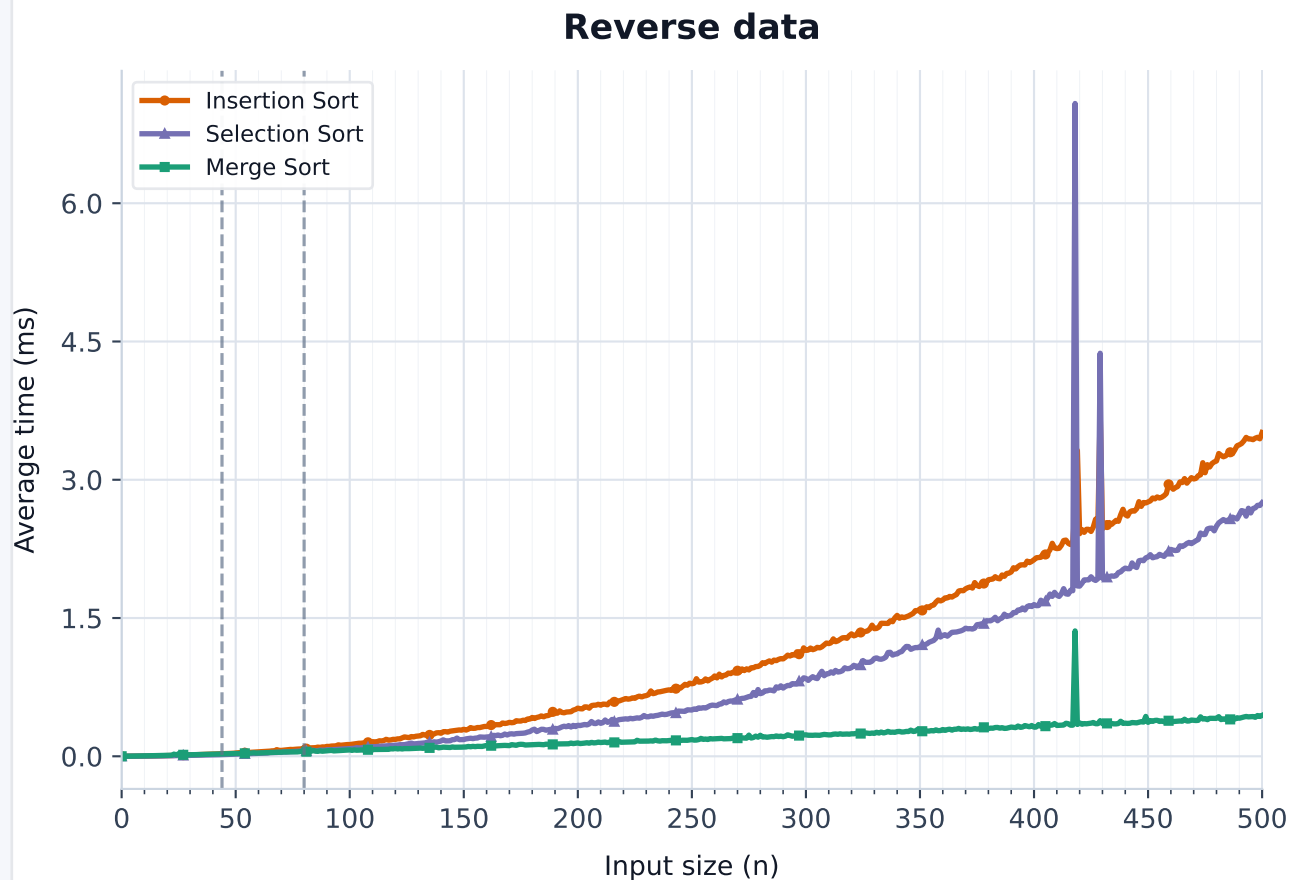
Selection vs Merge
 $n = 78$

Reading note

Lower line means faster.
Dashed line means stable crossover.

Reverse input

Average running time by input size



Key results

Fastest at largest n

Merge Sort
when n = 500

Stable crossovers

Insertion vs Merge
n = 44

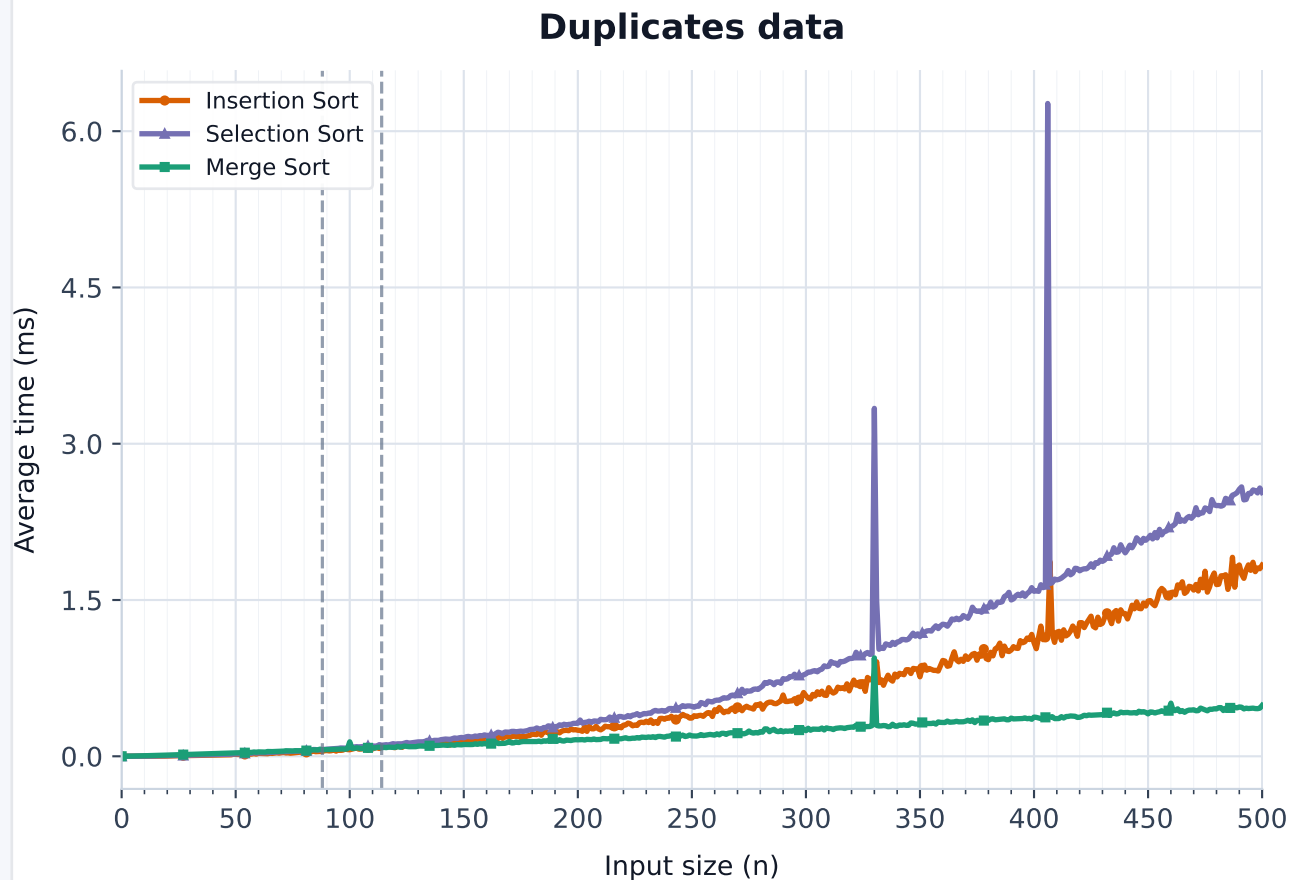
Selection vs Merge
n = 80

Reading note

Lower line means faster.
Dashed line means stable
crossover.

Duplicates input

Average running time by input size



Key results

Fastest at largest n

Merge Sort
when n = 500

Stable crossovers

Insertion vs Merge
n = 114

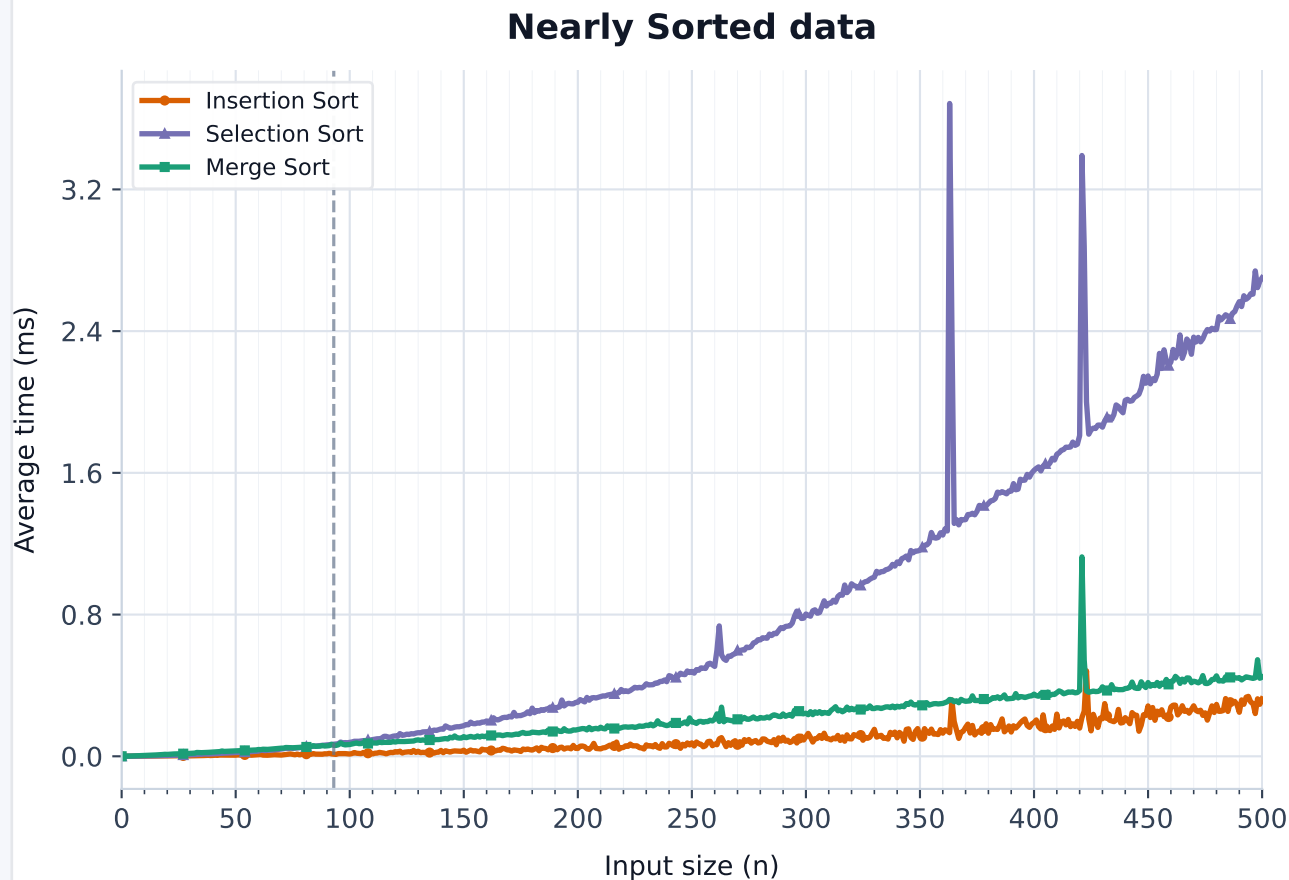
Selection vs Merge
n = 88

Reading note

Lower line means faster.
Dashed line means stable
crossover.

Nearly Sorted input

Average running time by input size



Key results

Fastest at largest n

Insertion Sort
when $n = 500$

Stable crossovers

Insertion vs Merge
No stable crossover

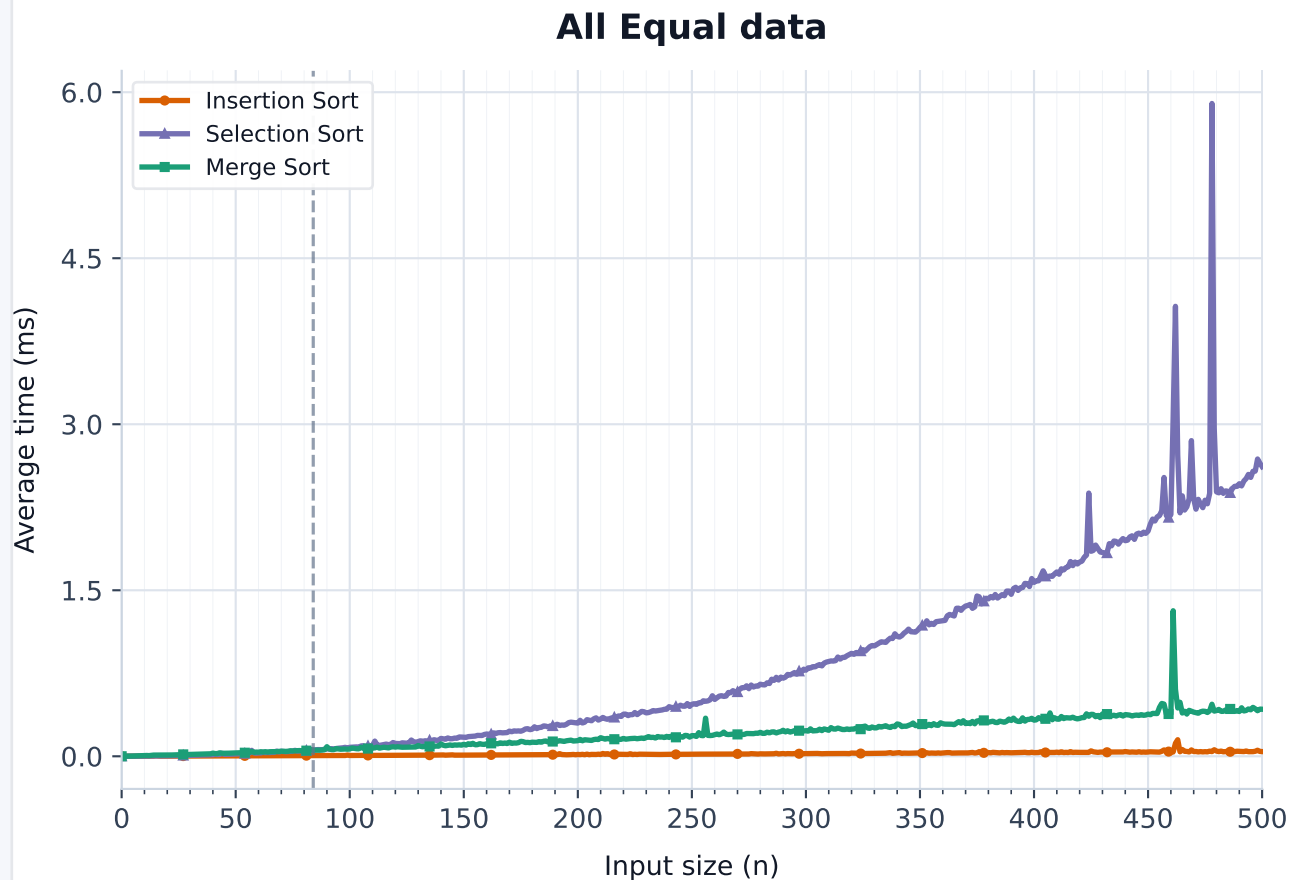
Selection vs Merge
 $n = 93$

Reading note

Lower line means faster.
Dashed line means stable crossover.

All Equal input

Average running time by input size



Key results

Fastest at largest n

Insertion Sort
when $n = 500$

Stable crossovers

Insertion vs Merge
No stable crossover

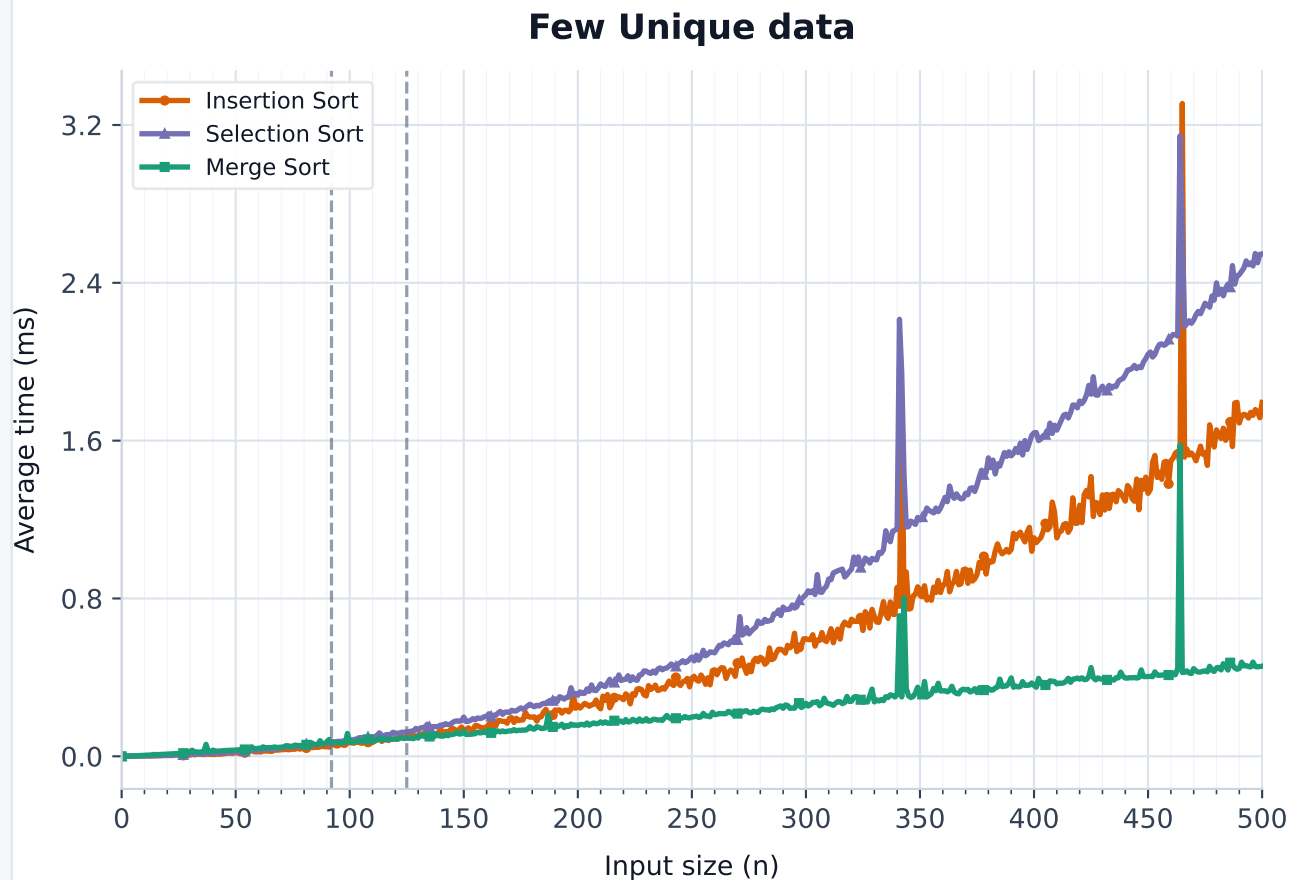
Selection vs Merge
 $n = 84$

Reading note

Lower line means faster.
Dashed line means stable
crossover.

Few Unique input

Average running time by input size



Key results

Fastest at largest n

Merge Sort
when $n = 500$

Stable crossovers

Insertion vs Merge
 $n = 125$

Selection vs Merge
 $n = 92$

Reading note

Lower line means faster.
Dashed line means stable crossover.